

merous issues in metering, smart metering and distribution automation applications, including magnetic interference/immunity challenges, EMI and others, all the way down to providing solid reference solutions to accelerate the design process.

A strong applications team is a big advantage, which sets a solution provider apart. Furthermore, you need a number of solutions (power supplies, RTCs, isolation products and transformer drivers) that surround the major system SoCs for various meter and smart grid systems. These aren't just off-the-shelf analogue components, rather they are high-performance analogue products developed for smart grid applications.

Could you give an example of how you were able to solve the EMI challenge in a smart grid application?

Many of these are proprietary to specific customers. One example, though, is from Germany, where meters must pass a test involving a massive magnet. This magnet is so strong that it immediately erases your credit cards and your computer hard drive. We had trouble even buying the magnet. In these types of cases, our applications engineers find signals buried deep within the noise. They make the solutions work, despite requirements that meters will likely never see in the field.

What functions can a system design engineer expect from next-gen SoC solutions for smart grid?

Our next highly integrated product is Zeus, which integrates our highest-performance metrology with hardware security and a 32-bit ARM core for applications and communication stacks. The security in this product is really differentiated because it is not just software security. We are implementing a lot of the features that we put into financial terminals.

What kinds of applications require a high-performance SoC, and what applications would be better off a lower-performance SoC to achieve pricing targets?

The ideal application for a high-performance SoC is a high-volume platform. If the platform needs to be leveraged across multiple regions, that's even better because firmware changes may be all that is needed. Lower-volume applications probably do better with performance building blocks or limited-function SoCs. If the region you are targeting has not selected a communication protocol, then it is probably safer to pick the building blocks.

How are the current smart grid setups made secure? How can design engineers improve upon the security features when using the new SoCs?

There are a variety of approaches in smart meter security today. The most common is tamper protection on the meter, followed by AES-128 encryption on the data being communicated to and from the meters. In this type of configuration, there are still gaps, such as the data path from the metrology to the communication hardware. If those lines can be sniffed or modified, then security has been breached.

Our recommendation is to secure the life-cycle of the product. Purchase through a trusted supply channel. Ensure authentication in the manufacturing process. This can be done with integrated or discrete products. Ensure authentication during the installation process as well. Finally, close gaps on the meter itself. For the example above, either select an integrated solution or bury the lines in the PCB and secure them separately, prior to routing to the communication module.

These solutions are much more difficult than this discussion warrants, but security is absolutely critical to our energy infrastructure and we need to build it into these systems, rather than patch it up later. That is definitely the best approach. ●

(The complete interview can be read at electronicsforu.com)